

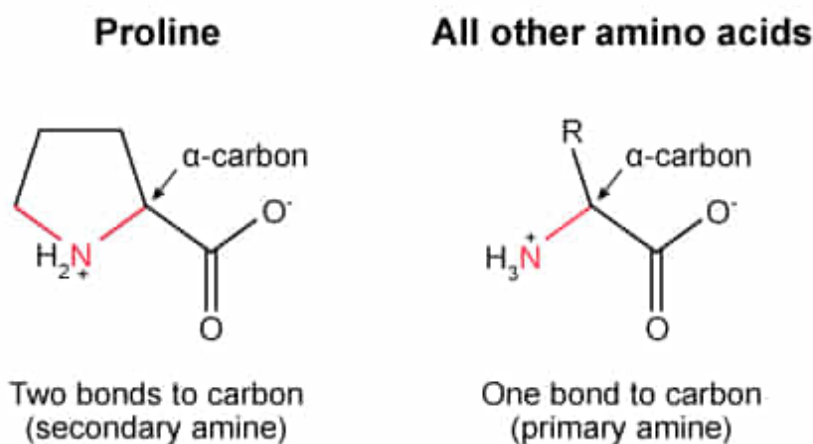
Proline differs from the other standard amino acids in that only proline:

- ☐ A. has a negative charge in its side chain.
- ☐ B. forms peptide bonds that are planar. [14%]
- ✔ ☒ C. contains a secondary amine in its backbone. [54%]
- ☐ D. is commonly found in flexible loop regions of folded proteins. [30%]

Correct

54% answered correctly

Explanation:



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The core structure of amino acids, called the **backbone**, consists of an **amino group** linked to an alkyl carbon (the **α-carbon**), which is linked to a

carboxyl group. The α -carbon is also linked to an **R group**, also known as the **side chain**, which varies from one amino acid to another and determines the identity of each amino acid. **Twenty standard amino acids**, each with a unique R group, are found in proteins in all domains of life.

Proline is unique among the standard amino acids because its R group forms a ring that includes the backbone amino group. Because of this unique feature, the amino group of the proline backbone has two bonds to carbon, making it a **secondary amine**. Although other amino acids (arginine, tryptophan, histidine) also have nitrogens bonded to two carbons in their *side chains*, **only proline** contains a **secondary amine** in its **backbone**.

(Choice A) The **acidic amino acids** (aspartate and glutamate) have negatively charged side chains.

Proline does not.

(Choice B) Peptide bonds are amide bonds. **Due to resonance**, all amide bonds (including peptide bonds) have partial double-bond character; therefore, all peptide bonds are planar, not just the bond from proline.

(Choice D) The R group of proline is bonded to the backbone at two points instead of one, making it relatively rigid and unlikely to be found in the flexible regions of a protein. Notably, β -turns, which commonly contain proline, are highly structured and inflexible. In addition, *many* amino acids are found in flexible-loop regions, so even if proline were common in these regions, its inclusion would not be a *unique* property of proline.

Educational objective:

All amino acids have a backbone consisting of an amino group, an α -carbon, and a carboxyl group. Each amino acid also has an R group that determines its identity. Proline is unique among the standard amino acids because its R group bonds to the backbone amino group, making it a secondary rather than a primary amine.

Time spent:0 sec

Subject:

Biochemistry

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System:

Amino Acids and Proteins